

Hitaansh Jain

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Education

New York University

GPA: 3.93/4.0 | BA, Computer Science, Minor: Mathematics

- Courses: Algorithm Problem Solving, Software Engineering, Computer Organization, Linear Algebra, Artificial Intelligence

New York, NY

Expected May 2027

Occidental College

Computer Science and Mathematics (transferred to NYU)

Los Angeles, CA

August 2024 – May 2025

Professional Experience

Header (joinheader.com)

Seattle, WA

Software Engineer Intern

May 2026 – August 2026

- Built the Follow-ups feature, letting users capture ideas from their briefings and follow them through to a recorded verdict: a 6-state lifecycle, 9 REST endpoints, 3 migrations, email reminders, and a React Native web UI, live in production.
- Shipped full-stack briefing customization to production (reorder, toggles, detail control, LLM TL;DR) by refactoring a hardcoded prompt into blocks assembled from a JSONB layout model, and cut TL;DR length by 45% with a word budget.
- Delivered the briefing formats and history view for Clear Tabs: a submit-time format picker with validated user-written sections, plus a React Native history screen on an authenticated, paginated FastAPI endpoint.
- Engineered an API that lets AI coding agents drive Follow-ups concurrently with their users: row-locked, forward-only sync with an audit event per change, credential-derived verdict authority, per-key approval gating, and a compare-and-set fix for a delete race.
- Wrote 750+ automated tests across an LLM pipeline, FastAPI/PostgreSQL backend, and React Native/Expo frontend, and merged every change through senior engineer review.

MathGPT (mathgpt.ai)

Remote

Software Engineer Intern, AI (Part-Time)

May 2026 – August 2026

- Designed a generator/critic verification architecture where the critic re-solves each problem in a fresh context before the generator's drafts exist and halts the run on mismatch, with Zod contracts and validator-error retries on the JSON artifacts and a LaTeX compile gate on the worksheet.
- Cut comparable run costs by 39% and runtime from 14.8 to 7.5 minutes via per-stage reasoning budgets and prompt caching, on a five-stage TypeScript/Next.js pipeline turning a calculus problem into a LaTeX case study, flashcards, and a walkthrough.
- Owned the corpus extraction system (45 OpenStax sections, 195 learning objectives) and a nine-table MySQL cache, covered by 1,000+ automated tests including negative controls, on a three-intern Agile team.

Vardhman Infotech

Remote

Software Engineer Intern (Part-Time)

December 2024 – May 2025

- Executed a full MySQL-to-Access migration for a Python inventory system serving 1,500+ retail locations to meet the client's offline, zero-infrastructure constraint: a 394-column schema across 19 tables, 700+ queries across 43 files, and a 73-function database layer refactored into a metadata-driven architecture.

Projects

Local RAG Search Platform

July 2025 – Present

- Built a local RAG system (FastAPI, Ollama, ChromaDB, React) over 1,070 pages of zoning codes with OCR, table-aware extraction, and network-isolated model and API containers, verifying each cited claim against its source and abstaining when support is missing.
- Built BM25-weighted hybrid retrieval and an evaluation harness for retrieval, answer, and citation-support accuracy, raising page-level hit@3 from 64% with vector search alone to 89% on 28 excerpt-verified questions.

Stock Analyzer

April 2026 – June 2026

- Built and deployed a 3-tier stock analysis platform (Docker Compose, Flask, MongoDB) with rate-limit-aware retries for the OpenAI and SEC APIs, an 8-worker job pipeline, and 80% coverage gates in GitHub Actions CI/CD.
- Modeled MongoDB data across 6 collections with upserts ensuring zero data loss across concurrent jobs, and built session-based auth plus a polling job monitor with a live log view.

VocabLearn

February 2026 – May 2026

- Led backend on a 5-person Agile team, delivering a JWT-authenticated Node.js/Express REST API with 13 endpoints and 21 integration tests, plus Gemini 2.5 Flash with structured JSON output for reverse-dictionary search-by-meaning.
- Engineered a quiz engine with distractor sets drawn from saved definitions and dual-layer deduplication (server-side Map plus client sessionStorage), gating all merges on 21 passing tests via a 3-workflow GitHub Actions pipeline.

Swordfight (Unity, C#) | Gameplay & Systems Programmer

September 2023 – May 2024

- Shipped a 14-character top-down fighting game to itch.io on a 10-person student team, drawing 150k page views and 4.5 stars across 111 ratings.
- Owned the polymorphic weapon architecture and HFSM enemy AI behind multi-stage boss fights, and ran the PlasticSCM branch strategy that kept unmergeable Unity scenes conflict-free.

Computer Skills

Languages & Frameworks: Python (FastAPI, Flask), TypeScript (Next.js, React), JavaScript (Node.js, Express, React Native/Expo), Java, C, C#, SQL (PostgreSQL, MySQL), MongoDB

Backend & DevOps: REST APIs, JWT/session auth, schema migrations, Docker, GitHub Actions CI/CD, Pytest, Vitest, Mocha, Claude Code

AI & Data Engineering: LLM pipelines (OpenAI, Gemini, Ollama, generator/critic verification), RAG, LangChain, ChromaDB, Pandas